

Q30 At time t , there are equal number of nuclides N_0 of X and Y.

After one half-life, number of X nuclide $N_X = N_0/2$

number of Y nuclide $N_Y = N_0/2 + N_0 = 3N_0/2$

i.e. the ratio of $N_X : N_Y = 1 : 3$

Answer: A